Please check the examination details below before entering your candidate information

Candidate surname	Other names

Centre Number | Candidate Number

Pearson Edexcel Level 3 GCE

Friday 15 May 2026

Afternoon

Paper reference **8FM0/27**

Further Mathematics

Advanced Subsidiary
Further Mathematics options
27: Decision Mathematics 1
(Part of opions D, F, H and K)

D1 Answer Book
Do not return the question paper with the answer book.

Total Marks

Turn over ▶

P81287A

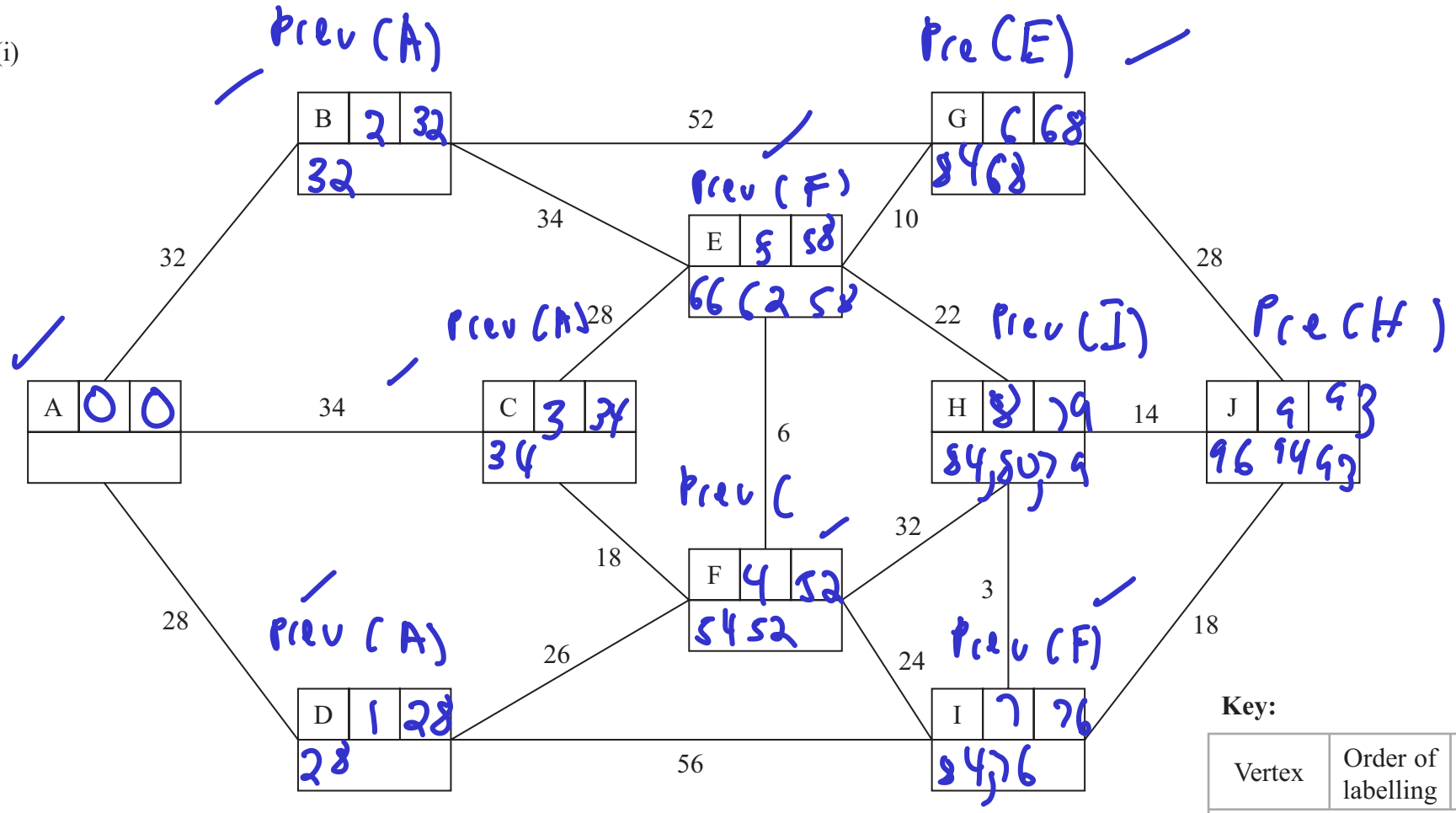
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P 8 1 2 8 7 A 0 1 1 2

Pearson

1.

(i)



Key:

Vertex	Order of labelling	Final value
Working values		

H → I → F → C → A

Figure 1

(a) Shortest time from A to J: 93

(b) Quickest route from A to J: A, C, F, I, H, J

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Question 1 continued

Time taken = (Scale factor)^2 × time

$= \left(\frac{6300}{20}\right)^2 \times 0.32$

16200 seconds

Sec → min

÷ 60

$\frac{16200}{60} = 270$ min

(Total for Question 1 is 8 marks)

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2.

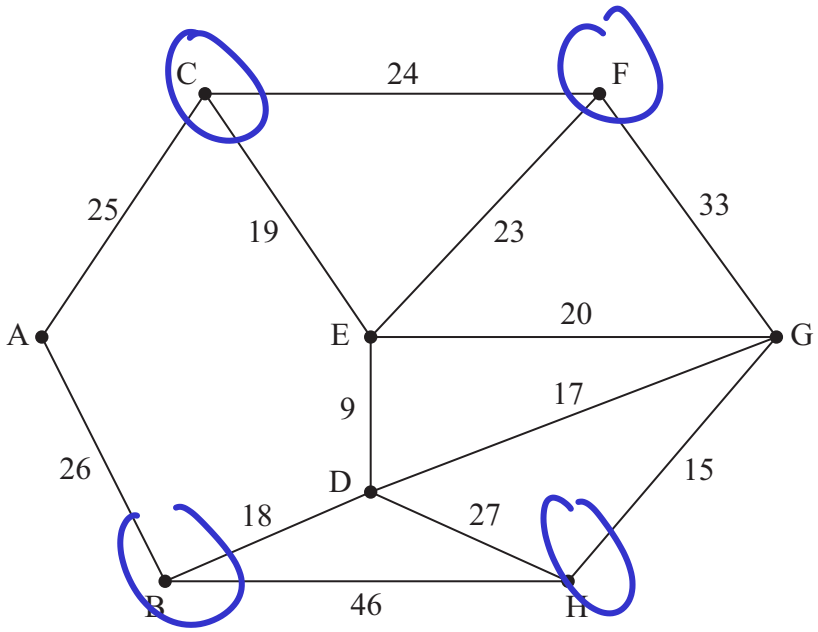


Figure 2

[The total weight of the network is 302]

Step 1 get the arcs in order

ED(9), HG(15), DG(17), BD(18)

(E(19), EG(20))

→ EF(23), (F(24),

(A(25), AB(26), DH(27)

FG(33)

Accepted

ED(9)

HG(15)

DG(17)

BD(18)

Rejected

EG(20)

CF(24)

AB(26

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CE (14)
EF (23)
CA (25)

DH (27)
FG (33)

(ii) full weight = 120

2b Cut is perpendicular bisectors are CJ, FJ, HJ, B

so pairs are CF + HB
CH + FB
CB + FH

CF + HCDB = 24 + 45 = 69°

C(ED)H + F(ED)B = 55 + 50 = 105

CDB + FH = 46 + 48 = 94

repeat CF, HP, BD

64 + 30 2 = 37 miles

(Total for Question 2 is 9 marks)

P 8 1 2 8 7 A 0 5 1 2

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3.

32 45 17 23 38 28 16 9 12 10

method = bubble sort

target = Descending order

pass 0 = 32, 45, 17, 23, 38, 28, 16, 9, 12, 10

pass 1 = 45, 32, 23, 38, 28, 17, 16, 12, 10, 9

pass 2 = 45, 32, 38, 28, 23, 17, 16, 12, 10, 9

pass 4 = 45, 38, 32, 28, 23, 17, 16, 12, 10, 9

Final pass: 45, 38, 32, 28, 23, 17, 16, 12, 10, 9

b

since 38 and 45 were not in the same bin

this implies that

n < 80

↳ no idea //

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Question 3 continued

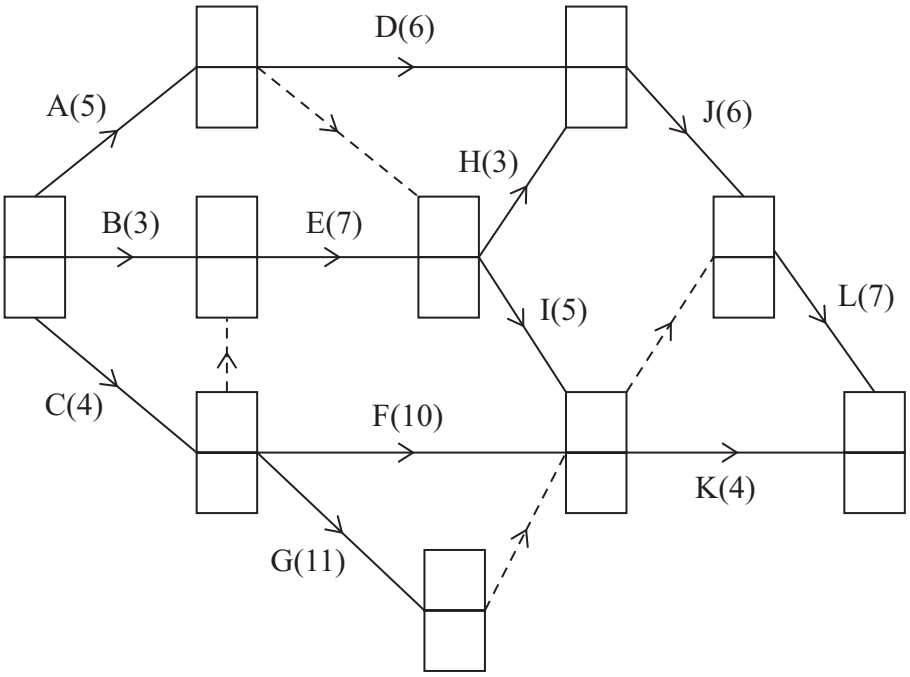
32 45 17 23 38 28 16 9 12 10

(Total for Question 3 is 6 marks)

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4.

Activity	Immediately preceding activities	Activity	Immediately preceding activities
A	–	G	C
B	–	H	
C	–	I	
D	A	J	
E	B, C	K	
F	C	L	



Key:

Early event time
Late event time

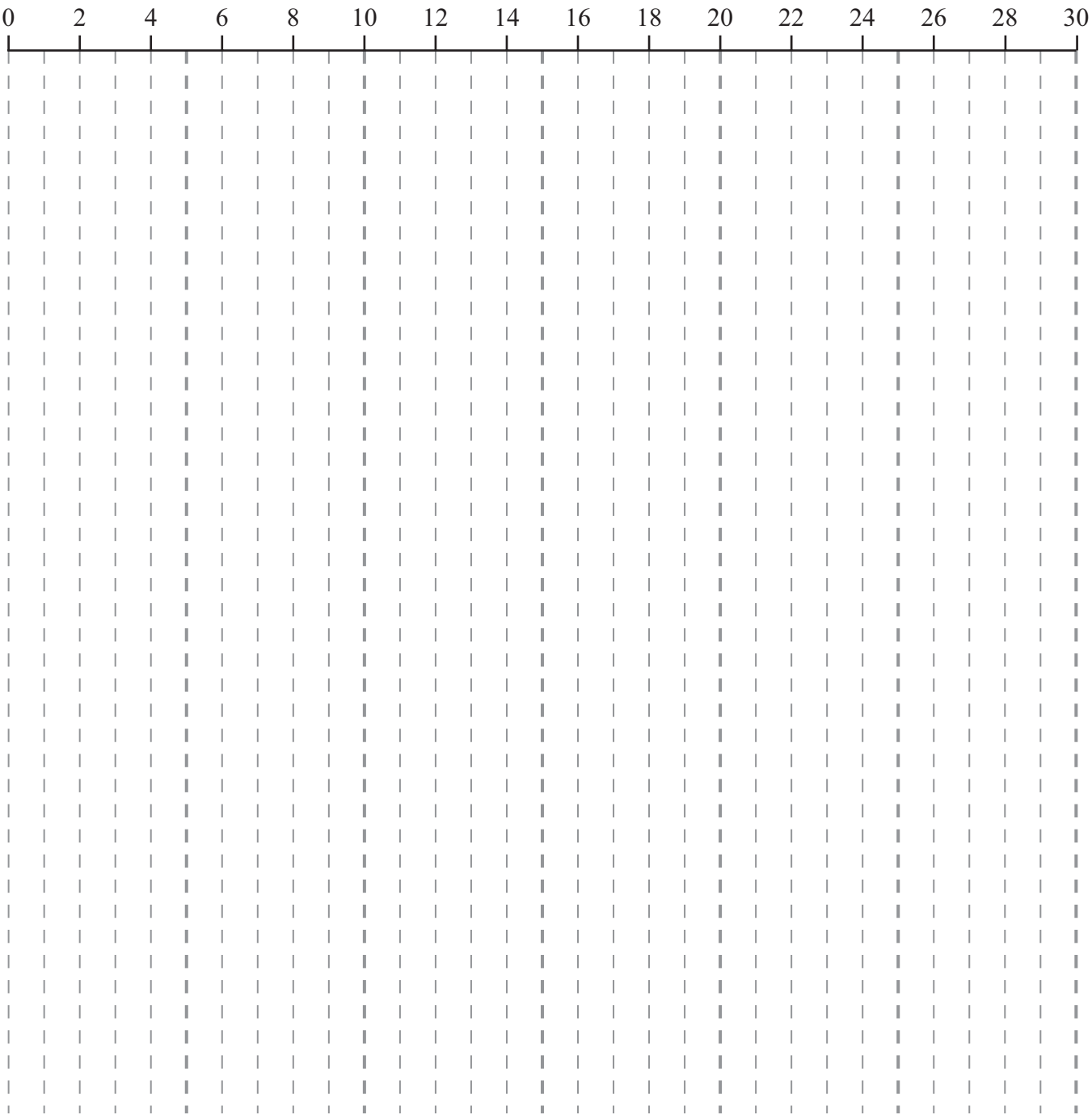
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Question 4 continued



(Total for Question 4 is 10 marks)

5.

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Question 5 continued

Question 5 continued

(Total for Question 5 is 7 marks)

TOTAL FOR DECISION MATHEMATICS 1 IS 40 MARKS

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